Quiz: Dynamo

Test your knowledge of Dynamo by completing this quiz.

Quiz ?

Statement

In CAP theorem terms, Dynamo is ____ system?

Select Multiple Options

Correct ✓
☑ AP (Available and Partition Tolerant)

False ⊗
☐ CP (Consistent and Partition Tolerant)

False ⊗
☐ CA (Consistent and Available)

Check Answer

Correct Answer ✓

Quiz ?

Statement

Amazon Dynamo is called a zero-hop DHT because:

Select Multiple Options

False ⊗
☐ A collision-free hash function is used.

False ⊗
☐ Consistent hashing allows the system to grow.

Correct ✓
☑ Clients know the complete set of servers and can directly contact the server that holds the data they need.

False ⊗
☐ Each server node maintains a finger table that identifies other nodes.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Does Dynamo always write data to the first 'N' nodes in the preference list?

Select Option

○ True

Correct ✓
● False

Explanation

Dynamo replicates writes to a sloppy quorum of other nodes in the system (which are the first N **healthy** nodes from the preference list, instead of a strict majority quorum).

Check Answer

Quiz ?

Statement

How does Dynamo resolve conflicts between different versions of data?

Select Multiple Options

False ⊗
☐ Merkle tree

False ⊗
☐ Hinted handoff

Correct ✓
☑ Vector clock

False ⊗
☐ Gossip protocol

Check Answer

Correct Answer ✓

Quiz ?

Statement

What strategies can be used to resolve data conflicts? Select all that apply.

Select Multiple Options

False ⊗
☐ Last write wins

False ⊗
☐ Conflict-free replicated data types (CRDTs).

False ⊗
☐ Sloppy quorum

Correct ✓
☑ Vector clocks

Check Answer

Correct Answer ✓

Quiz ?

Statement

How does Dynamo handle temporary node failures? Select all that apply.

Select Multiple Options

False ⊗
☐ Merkle tree

Correct ✓
☑ Hinted handoff

Correct ✓
☑ Sloppy quorum

False ⊗
☐ Gossip protocol

Check Answer

Correct Answer ✓

Quiz ?

Statement

What are the benefits of Vnodes in Dynamo? Select all that apply.

Select Multiple Options

Correct ✓
☑ Vnodes help spread the load more evenly across the physical nodes on the cluster. This speeds up the rebalancing process after adding or removing nodes.

False ⊗
☐ Vnodes enable the use of virtual machines to help Dynamo scale better.

Correct ✓
☑ Vnodes make it easier to maintain a cluster containing heterogeneous machines.

Correct ✓
☑ Use of Vnodes decreases the probability of hotspots, as Vnodes help assign smaller ranges to each physical node.

Check Answer

Correct Answer ✓

Quiz ?

Statement

What is optimistic replication in Dynamo?

Select Multiple Options

False ⊗
☐ When replicating, Dynamo chooses the closest nodes to the coordinator node to store the replicas.

False ⊗
☐ Dynamo tries to distribute replicas on different racks.

Correct ✓
☑ Replicas may diverge and are not guaranteed to be identical at all times.

False ⊗
☐ All replicas are updated atomically, ensuring ACID semantics.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Dynamo uses Vector clocks to:

Select Multiple Options

False ⊗
☐ Impose a total ordering of all get and put operations in the system.

False ⊗
☐ Disallow concurrent updates of the same object by choosing the earliest writer.

False ⊗
☐ Determine the sequence of write operations across multiple objects.

Correct ✓
☑ Identify conflicting versions of the same object stored on different servers.

Check Answer

Correct Answer ✓

Quiz ?

Statement

Dynamo optimizes key lookups via:

Select Multiple Options

False ⊗
☐ Consistent hashing

False ⊗
☐ Finger tables

Correct ✓
☑ Storing the table of all nodes in the system at each node.

False ⊗
☐ Storing all key-value sets in a hash table at each node.

Check Answer

Correct Answer ✓

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☐ Mark as Completed